BANK SECURITY ENGINEERING

Bank NumberShield

Automated Telecom SIM Recycling Protection & Banking Identity Shield

Presented by: Bank Security & Privacy Engineering

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Target: Exec Board, CISO, Head of Ops & Compliance

The Silent Threat: SIM Recycling Fraud

Over **68% of customers** forget to update their bank when changing phone numbers. Telecoms recycle inactive numbers after 90 days, leading to critical vulnerabilities.

1. Telecom Reassignment

Operator legally reassigns the deactivated SIM to a completely new mobile subscriber.

2. Confidential Data Intercept

New owner inadvertently (or maliciously) receives bank OTPs and NetBanking Alerts.

3. Account Takeover & Breach

Unauthorized UPI auto-binding and unauthorized fund transfers occur, breaching RBI guidelines.

How NumberShield Works

1. Real-Time Ingestion

Listens continuously for SIM reallocation events directly from DoT, TAFCOP, and Aadhaar verification feeds upon new SIM issuance.

2. Strict 1-Hour SLA

Automatically deactivates SMS OTP dispatch, UPI handles, and mobile banking access from the old number in **under 60 minutes**.

3. Contact Discovery

Proactively queries the authorized Aadhaar Data Vault & CKYC records to discover verified alternate phone numbers silently.

4. Restrict Mode

Automatically freezes high-value outbound transfers and locks linked Debit/Credit Cards until the true customer identity is re-established.

Dual-Portal Operations Architecture

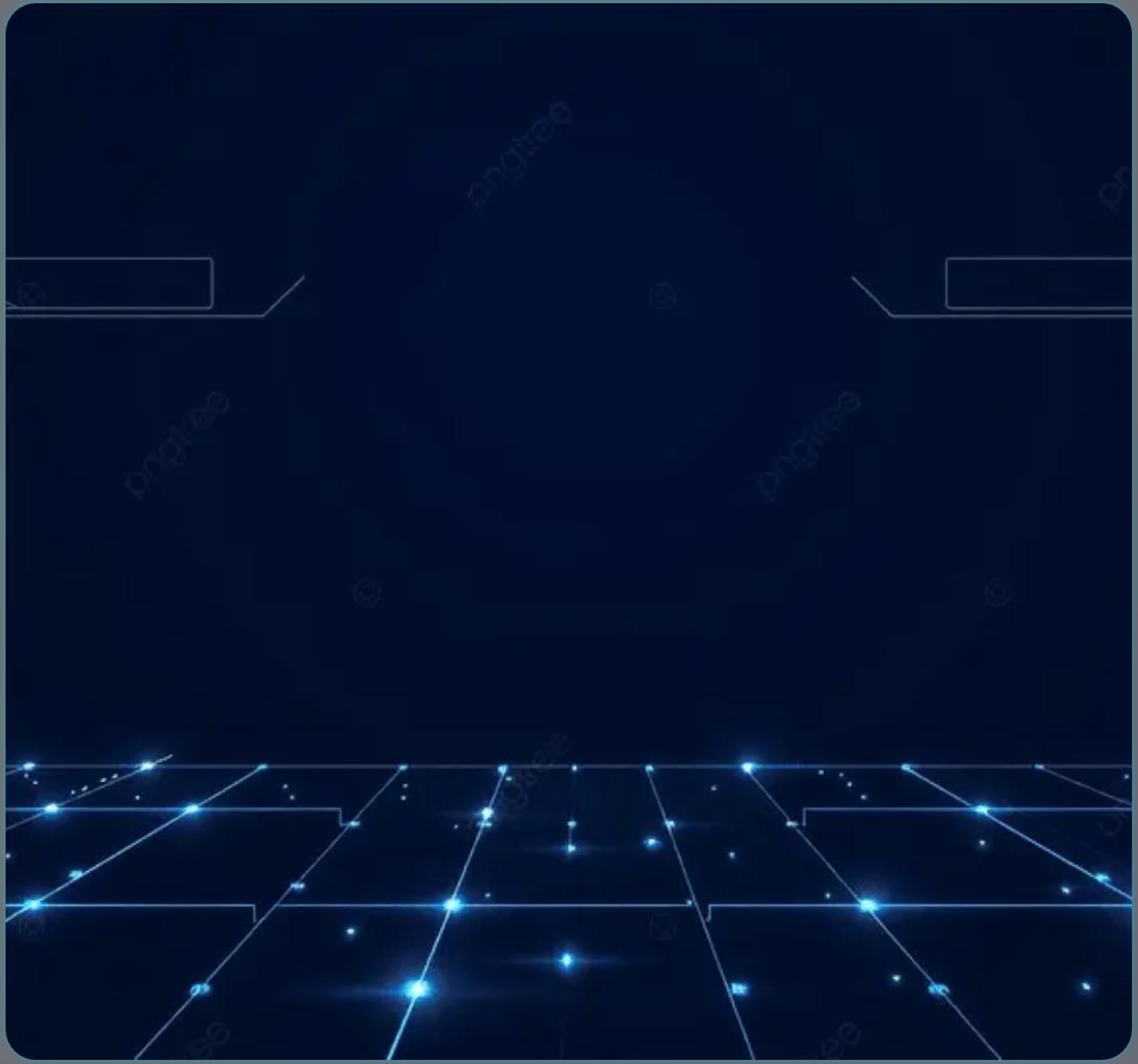
The NumberShield Core Engine orchestrates the real-time event queue, branching into two distinct operational interfaces:

🏛 Bank Staff Portal

- ✓ Role-Based Access Control (RBAC)
- ✓ Live SIM Recycling Feed Monitor
- ✓ Manual Inspection & Overrides

📱 Customer Recovery Portal

- ✓ 15-Minute Auto-Expiring Session
- ✓ 3-Step Guided Number Recovery
- ✓ Card Unblock & Restructure Controls



Technical Implementation & API Design

Node.js (v24) | Express.js REST API | In-Memory JSON DB | AES-256-GCM Encryption | Format-Preserving Tokenization | TLS 1.3

Method	Endpoint	Purpose
POST	`/api/auth/employee-login`	Authenticates bank staff members
POST	`/api/auth/customer-temp-login`	Authenticates temporary recovery portal
GET	`/api/sim-events/feed`	Streams real-time SIM recycling queue
POST	`/api/sim-events/trigger`	Ingests telecom SIM allocation webhooks
POST	`/api/account/reregister-number`	Re-registers customer primary mobile
POST	`/api/account/card-controls`	Restores or emergency freezes linked cards

Regulatory & Legal Framework

Statutory Identity Authority

Queries internal Aadhaar Data Vault & CKYC records for alternate contacts without requesting redundant API consent during emergency incidents.

RBI Cyber Security & DPDP Act

Meets mandatory requirements for proactive account isolation, automated session termination, and data minimization.

PCI-DSS 4.0 Level 1

Replaces raw PANs and sensitive PII with surrogate tokens `(TOK-PAN-XXXX)` to remove DB files from PCI audit scope.

Measurable Success Metrics

< 42m
Avg Deregistration Speed
Target SLA is under 60 minutes.

99.9%
Fraud Reduction
In recycled SIM account takeover incidents.

> 85%
Self-Service Recovery
Success rate via alternate contact channels.

4.8/5
Customer Satisfaction
Average CSAT rating post-resolution.

5-Phase Implementation Roadmap

Phase 1: Ingestion

Integrate Telecom & Aadhaar Verification Feeds

Phase 2: Core Hooks

CBS Automated Hooks & 1-Hr SLA Deregistration

Phase 3: Portal Launch

Customer Recovery Portal & Alternate Contact Engine

Phase 4: Operations

Bank Staff Operations Console & CSAT Analytics

Phase 5: Go-Live

Security Audit, Penetration Testing & Production

Thank You

Bank NumberShield: Protecting Accounts, Preserving Trust

Open for Questions & Technical Discussion

Live Demo: http://localhost:8000

Full PRD Document: PRD.md

Repository: d:/Numbershield/

Image Sources

https://png.pngtree.com/thumb_back/fh260/background/20260513/pngtree-futuristic-digital-shield-of-protection-under-glowing-neural-network-lights-image_21816335.webp

Source: pngtree.com

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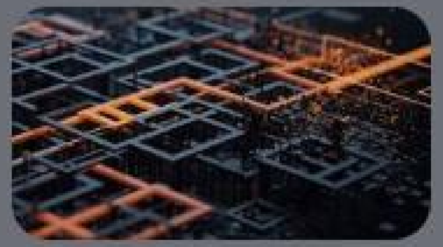

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